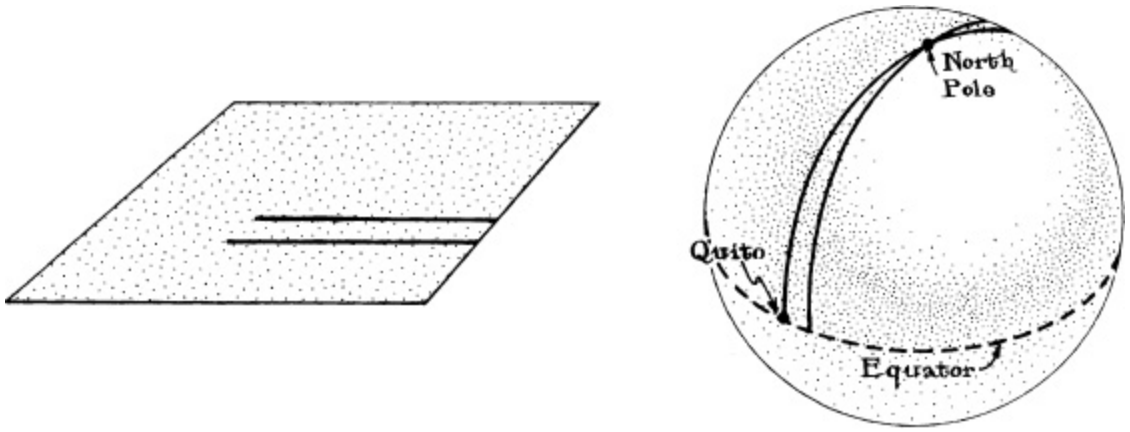


2.4 Two straight lines, initially parallel, never cross on a flat surface such as the sheet of paper shown on the left. Two straight lines, initially parallel, will typically cross on a curved surface such as the globe of the world shown on the right.

The second sense of straightness is the one that we shall use below, when discussing spacetime: In sufficiently small regions on the globe along the great circle's route, the globe's curvature can hardly be noticed. In such a region, the great circle looks straight in the usual flat-sheet-of-paper sense of straightness—the sense of straightness used by professional surveyors, who lay out boundaries of property using transits or laser beams. The great circle is straight, in this surveyors' sense, in each and every small region along its route.

Mathematicians use the name *geodesic* for any line, on a curved or warped surface, that is straight in these two senses: the airlines' "shortest route" sense, and the surveyors' sense.

Now move eastward on the globe from Quito by a few centimeters, and construct a new straight line (great circle; geodesic) that is precisely parallel, at the equator, to the one through Quito. This straight line, like the first one, will pass through the globe's North Pole. *It is the curvature of the globe's surface that forces the two straight lines, initially parallel to cross at the North Pole.*

With this understanding of the effects of curvature in two-dimensional surfaces, we can return to four-dimensional spacetime and ask about curvature there.

In an idealized universe without gravity, there is no warpage of space, no warpage of time; spacetime has no curvature. In such a universe,